



AIOps: Transforming IT Operations with Generative AI

AIOps solution with Kiro CLI and MCP Server

Dasapich Thongnopnua

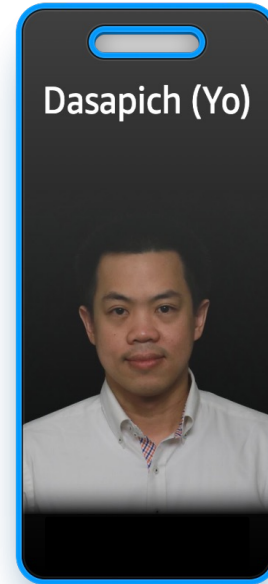
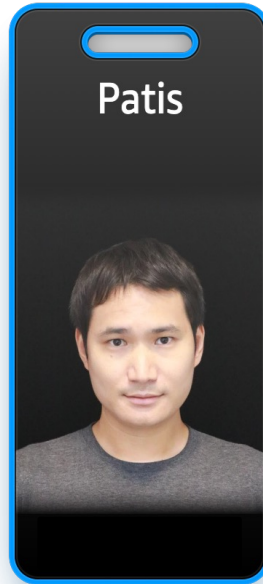
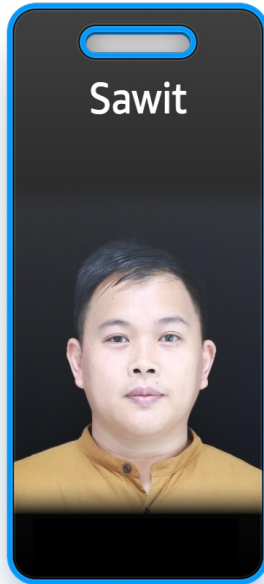
Senior Solutions Architect
AWS

Sawit Meekwamdee

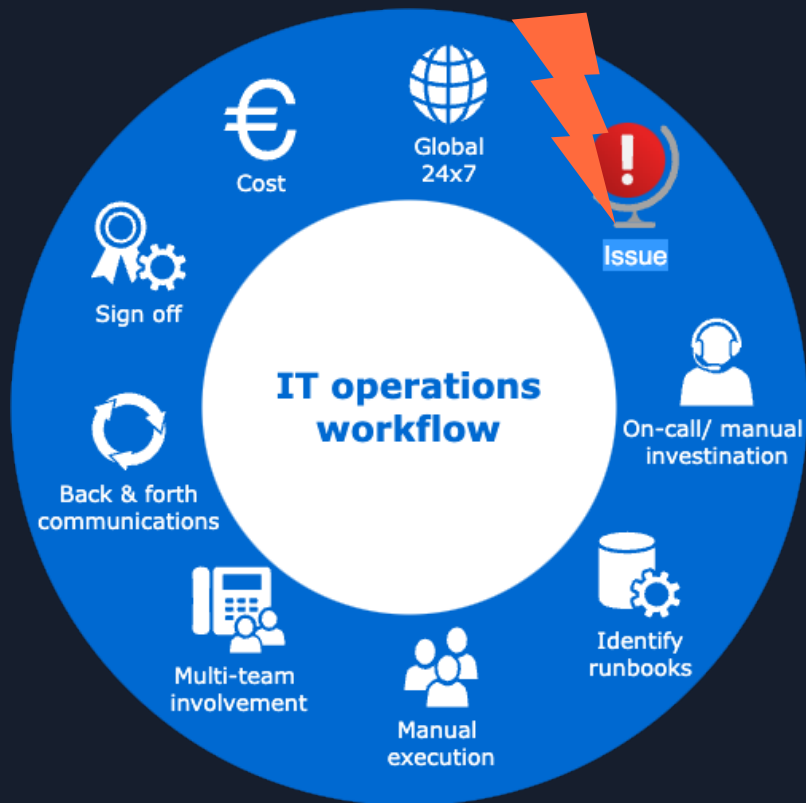
Solutions Architect
AWS

Patis Piriayahaphan

Solutions Architect
AWS



What IT operation lifecycle **looks like**



Longer resolution time

Prone to human errors

Dependency on skilled engineers

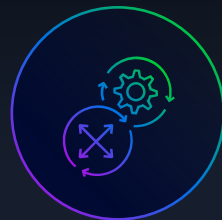
IT operations challenges



Runbook dependency



Manual diagnosis



Lack of automated
remediation process



Missing real time
visibility

KIRO



Helps developers and IT professionals build faster across the entire software development lifecycle (SDLC)

Most accurate coding recommendations

Agents can autonomously help you implement features, refactor code, perform software upgrades and more

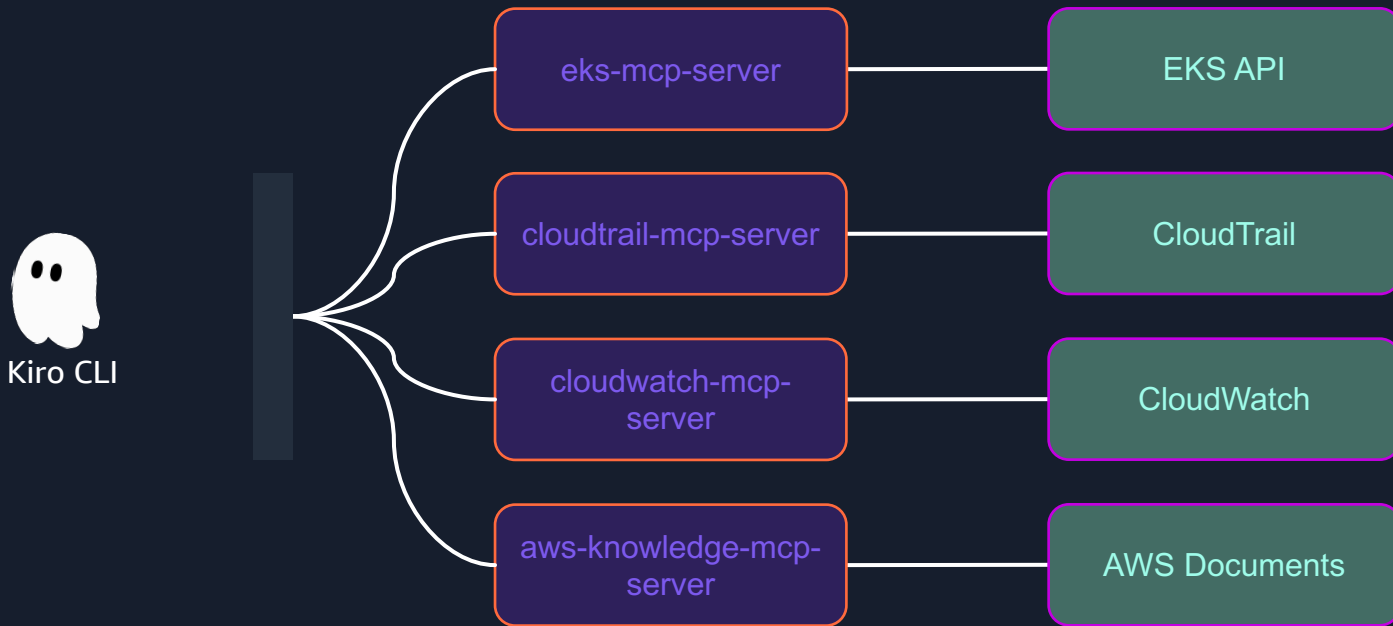
KIRO is an expert on AWS and optimizing AWS environment

Best-in-class security vulnerability scanning and remediation

Kiro is built with security and privacy in mind from the start, making it easier for organizations to use generative AI safely.



Leveraging MCP



Introducing the EKS MCP Server

Empower AI agents with
resource management and
real-time cluster visibility



Kubernetes API

Allow LLMs to manage Kubernetes resources like Pods and Deployments



Troubleshooting

Access Kubernetes events and logs, search a knowledge base



Cluster management

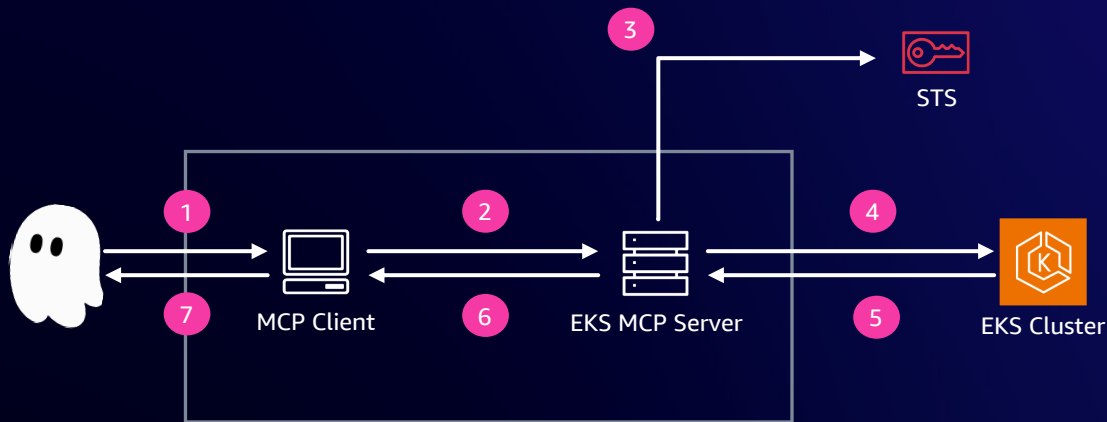
Manage simple EKS Auto Mode clusters through CloudFormation



Deploy applications

Help LLMs apply Kubernetes YAML manifests

EKS MCP Server – Troubleshooting



`list_k8s_resources`

`manage_k8s_resource`

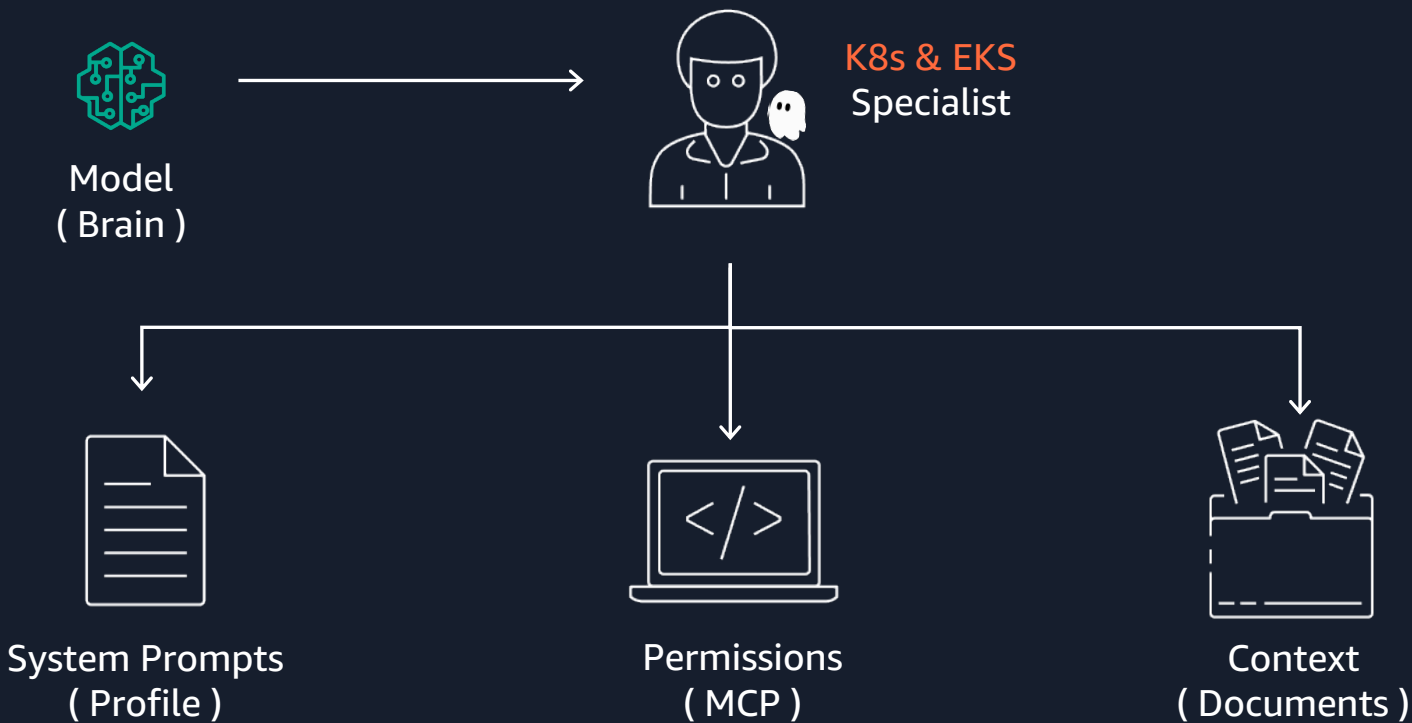
`list_api_versions`

`apply_yaml`

`get_pod_logs`

`get_k8s_events`

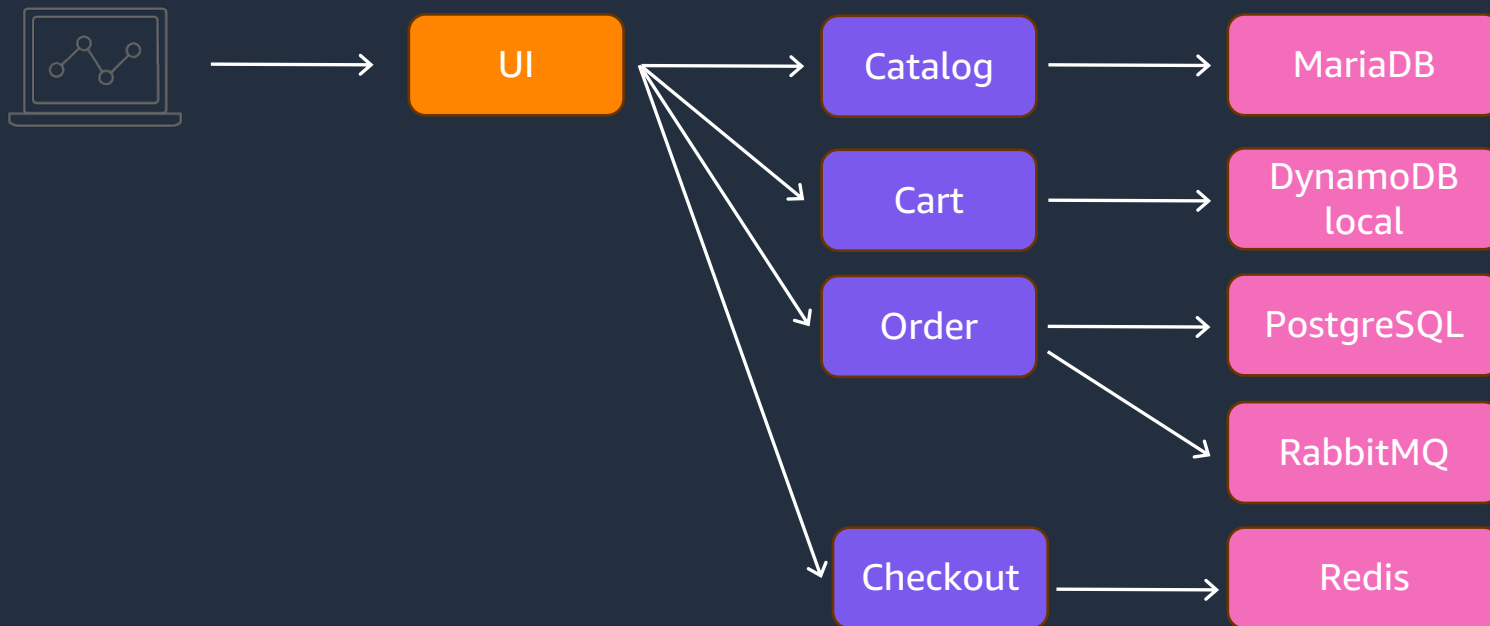
Specialized Agents



Demo Time !!!



Demo Retail Store



Shop for top products in our store

Latest Products, Low Prices

Hot products!

SAMPLE
PRODUCT

SAMPLE
PRODUCT

SAMPLE
PRODUCT

SAMPLE
PRODUCT

Ask KIRO-CLI for Help



Let's recap what we've shown



Fault Injected

Issue 1: Cart Service - Memory Pressure

- Config: Memory limit 192Mi (barely enough)
- Symptom: Intermittent slowness, occasional restarts

Issue 2: Catalog Service - Wrong DB Endpoint

- Config: DB_ENDPOINT=catalog-mysql-svc:3306 (should be catalog-mysql:3306)
- Symptom: Some products fail to load

Issue 3: Orders Service - Aggressive Probes

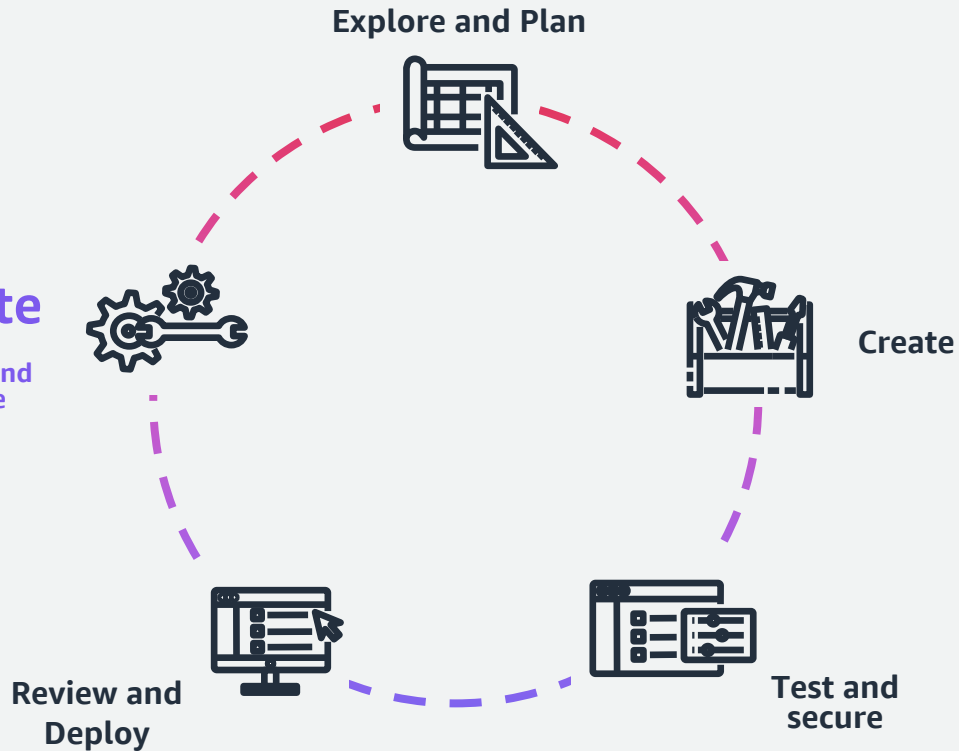
- Config: initialDelaySeconds=20, failureThreshold=2 (too aggressive)
- Symptom: Occasional order failures during startup

User-Visible Symptoms Observed

-  Homepage loads (UI is fine)
-  Product catalog sometimes shows errors
-  Adding to cart is slow and sometimes fails
-  Checkout process times out occasionally
-  Orders sometimes fail to save

Start leveraging
Kiro
for your entire
development
lifecycle today!

Operate
Maintain,
Transform and
Modernize



Get started with **Kiro** for free



Kiro.dev



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COMPUTING FOUNDATION

Thank You